

2.5 Moments of partitioned random vectors

Regression with random regressors studies a response vector $\mathbf{Y}$ ($p \times 1$) together with a regressor vector $\mathbf{X}$ ($q \times 1$) observed on the same unit. In this section $\mathbf{X}$ is a random vector, not a model matrix. We stack the two into one vector and read off what the blocks of its moments mean.

2.5.1 Blocks of the mean and covariance

Theorem 2.5.1 — Partitioned moments

Let $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$ have finite second moments, and write

$$\mathbf{E}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_X \end{pmatrix}, \quad \text{Cov}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\Sigma}_{YY} & \boldsymbol{\Sigma}_{YX} \\ \boldsymbol{\Sigma}_{XY} & \boldsymbol{\Sigma}_{XX} \end{pmatrix}.$$

- (a) $\boldsymbol{\mu}_Y = \mathbf{E}(\mathbf{Y})$, $\boldsymbol{\mu}_X = \mathbf{E}(\mathbf{X})$, $\boldsymbol{\Sigma}_{YY} = \text{Cov}(\mathbf{Y})$, $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$, $\boldsymbol{\Sigma}_{YX} = \text{Cov}(\mathbf{Y}, \mathbf{X})$ is $p \times q$, and $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{YX}^\top$.
- (b) For constants $\mathbf{A}$, $\mathbf{B}$, $\mathbf{c}$ of conformable sizes, $\text{Cov}(\mathbf{AY}, \mathbf{BX}) = \mathbf{A}\boldsymbol{\Sigma}_{YX}\mathbf{B}^\top$, and

$$\text{Cov}(\mathbf{AY} + \mathbf{BX} + \mathbf{c}) = \mathbf{A}\boldsymbol{\Sigma}_{YY}\mathbf{A}^\top + \mathbf{A}\boldsymbol{\Sigma}_{YX}\mathbf{B}^\top + \mathbf{B}\boldsymbol{\Sigma}_{XY}\mathbf{A}^\top + \mathbf{B}\boldsymbol{\Sigma}_{XX}\mathbf{B}^\top.$$

- (c) $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ and $\mathcal{C}(\boldsymbol{\Sigma}_{YX}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{YY})$. Equivalently, for every generalized inverse $\boldsymbol{\Sigma}_{XX}^-$,

$$\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XY}, \quad \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX},$$

and similarly with the roles of $\mathbf{Y}$ and $\mathbf{X}$ exchanged.

Proof.

- (a) The outer product $(\mathbf{W} - \mathbf{E}\mathbf{W})(\mathbf{W} - \mathbf{E}\mathbf{W})^\top$ has blocks $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top$, $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$, its transpose, and $(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$. Take expectations blockwise. (b) $\mathbf{AY} + \mathbf{BX} + \mathbf{c} = [\mathbf{A}, \mathbf{B}]\mathbf{W} + \mathbf{c}$, and $\mathbf{AY} = [\mathbf{A}, \mathbf{0}]\mathbf{W}$, $\mathbf{BX} = [\mathbf{0}, \mathbf{B}]\mathbf{W}$. Apply Theorem 2.2.1 and multiply out the blocks.
- (b) Let $\mathbf{a} \in \mathcal{N}(\boldsymbol{\Sigma}_{XX})$. By Theorem 2.2.3(c), $\mathbf{a}^\top \mathbf{X}$ is constant with probability one, so its covariance with $\mathbf{Y}$ vanishes: $\boldsymbol{\Sigma}_{YX}\mathbf{a} = \text{Cov}(\mathbf{Y}, \mathbf{a}^\top \mathbf{X}) = \mathbf{0}$. Thus $\mathcal{N}(\boldsymbol{\Sigma}_{XX}) \subseteq \mathcal{N}(\boldsymbol{\Sigma}_{YX})$. Taking orthogonal complements, and using $\mathcal{N}(\mathbf{B})^\perp = \mathcal{C}(\mathbf{B}^\top)$ and the symmetry of $\boldsymbol{\Sigma}_{XX}$, gives $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$. The same argument with $\mathbf{Y}$ and $\mathbf{X}$ exchanged gives the second inclusion. A column inclusion $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ means $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\mathbf{K}^\top$ for some $\mathbf{K}$. Then $\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}^\top = \boldsymbol{\Sigma}_{XY}$. Conversely, that identity exhibits every column of $\boldsymbol{\Sigma}_{XY}$ as a combination of columns of $\boldsymbol{\Sigma}_{XX}$ (Theorem 1.4.1). Transposing gives $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$ and the second identity. ■

Part (c) looks technical, but it removes every difficulty with singular $\boldsymbol{\Sigma}_{XX}$ in what follows. A direction in which $\mathbf{X}$ does not vary cannot covary with anything.

2.5.2 Linear prediction and residual covariance

The most important use of the blocks is to split $\mathbf{Y}$ into a part explained linearly by $\mathbf{X}$ and a part uncorrelated with $\mathbf{X}$.

Theorem 2.5.2 — Best linear predictor

In the setting of Theorem 2.5.1, fix a generalized inverse Σ_{XX}^- and put

$$\mathbf{B}_* = \Sigma_{YX} \Sigma_{XX}^-, \quad \mathbf{E} = \mathbf{Y} - \mu_Y - \mathbf{B}_*(\mathbf{X} - \mu_X), \quad \Sigma_{YY \cdot X} = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^- \Sigma_{XY}.$$

- (a) $E(\mathbf{E}) = \mathbf{0}$ and $\text{Cov}(\mathbf{E}, \mathbf{X}) = \mathbf{0}$.
- (b) $\text{Cov}(\mathbf{E}) = \Sigma_{YY \cdot X}$. This matrix is nonnegative definite and does not depend on the choice of generalized inverse. Neither does $\mathbf{B}_*(\mathbf{X} - \mu_X)$, with probability one.
- (c) For every constant $p \times q$ matrix $\mathbf{B}$ and $p \times 1$ vector $\mathbf{c}$,

$$E[(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})^\top] = \Sigma_{YY \cdot X} + (\mathbf{B} - \mathbf{B}_*)\Sigma_{XX}(\mathbf{B} - \mathbf{B}_*)^\top + \mathbf{d}\mathbf{d}^\top,$$

where $\mathbf{d} = \mu_Y - \mathbf{c} - \mathbf{B}\mu_X$. So the mean squared error matrix is never smaller, in the nonnegative definite ordering, than $\Sigma_{YY \cdot X}$. Equality holds iff $\mathbf{c} = \mu_Y - \mathbf{B}\mu_X$ and $\mathbf{B}\Sigma_{XX} = \Sigma_{YX}$.

Proof. By Theorem 2.5.1(c), write $\Sigma_{YX} = \mathbf{K}\Sigma_{XX}$. Then $\mathbf{B}_*\Sigma_{XX} = \mathbf{K}\Sigma_{XX}\Sigma_{XX}^-\Sigma_{XX} = \mathbf{K}\Sigma_{XX} = \Sigma_{YX}$.

- (a) The mean is clear. By Theorem 2.5.1(b), $\text{Cov}(\mathbf{E}, \mathbf{X}) = \Sigma_{YX} - \mathbf{B}_*\Sigma_{XX} = \mathbf{0}$.
- (b) Again by Theorem 2.5.1(b), $\text{Cov}(\mathbf{E}) = \Sigma_{YY} - \mathbf{B}_*\Sigma_{XY} - \Sigma_{YX}\mathbf{B}_*^\top + \mathbf{B}_*\Sigma_{XX}\mathbf{B}_*^\top$. Using $\mathbf{B}_*\Sigma_{XX} = \Sigma_{YX}$, the last term is $\Sigma_{YX}\mathbf{B}_*^\top$, which cancels the third, so $\text{Cov}(\mathbf{E}) = \Sigma_{YY} - \mathbf{B}_*\Sigma_{XY} = \Sigma_{YY \cdot X}$. Next, $\Sigma_{YX}\Sigma_{XX}^-\Sigma_{XY} = \mathbf{K}\Sigma_{XX}\Sigma_{XX}^-\Sigma_{XX}\mathbf{K}^\top = \mathbf{K}\Sigma_{XX}\mathbf{K}^\top$, which involves no generalized inverse. A covariance matrix is nonnegative definite. Finally, by Theorem 2.2.3(d), $\mathbf{X} - \mu_X = \Sigma_{XX}\mathbf{v}$ for some random $\mathbf{v}$, with probability one, and then $\mathbf{B}_*(\mathbf{X} - \mu_X) = \Sigma_{YX}\mathbf{v}$ for every choice of generalized inverse.
- (c) Decompose $\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X} = \mathbf{E} + (\mathbf{B}_* - \mathbf{B})(\mathbf{X} - \mu_X) + \mathbf{d}$. The three pieces are, respectively, mean zero, mean zero and constant, and the first two are uncorrelated by (a). So the cross products have zero expectation, and the expected outer product is $\text{Cov}(\mathbf{E}) + (\mathbf{B}_* - \mathbf{B})\Sigma_{XX}(\mathbf{B}_* - \mathbf{B})^\top + \mathbf{d}\mathbf{d}^\top$. Both added matrices are nonnegative definite. They vanish iff $\mathbf{d} = \mathbf{0}$ and $(\mathbf{B} - \mathbf{B}_*)\Sigma_{XX}^{1/2} = \mathbf{0}$, and the latter is equivalent to $(\mathbf{B} - \mathbf{B}_*)\Sigma_{XX} = \mathbf{0}$, that is, to $\mathbf{B}\Sigma_{XX} = \Sigma_{YX}$. ■

The affine function $\mu_Y + \mathbf{B}_*(\mathbf{X} - \mu_X)$ is the **best linear predictor** of $\mathbf{Y}$ from $\mathbf{X}$. Because the ordering in (c) is the nonnegative definite ordering, it is simultaneously best for every linear combination $\mathbf{a}^\top \mathbf{Y}$ and for the total squared error. The matrix $\Sigma_{YY \cdot X}$ is the **partial covariance matrix** of $\mathbf{Y}$ given $\mathbf{X}$. When Σ_{XX} is nonsingular, $\mathbf{B}_* = \Sigma_{YX}\Sigma_{XX}^{-1}$ holds the population regression coefficients, and $\Sigma_{YY \cdot X}$ is a Schur complement. Two consequences follow from Chapter 1. If $\text{Cov}(\mathbf{W})$ is positive definite, the YY block of its inverse is $\Sigma_{YY \cdot X}^{-1}$ (Theorem 1.6.4). And $\det \text{Cov}(\mathbf{W}) = \det(\Sigma_{XX}) \det(\Sigma_{YY \cdot X})$ (Theorem 1.6.3), so the generalized variance factorizes

into the part of $\mathbf{X}$ and the part of $\mathbf{Y}$ that $\mathbf{X}$ leaves unexplained.

Nothing here says that $E(\mathbf{Y} | \mathbf{X})$ is linear. The best linear predictor is best among affine functions, whatever the joint distribution. For jointly normal vectors, the conditional mean happens to be this affine function, and $\Sigma_{Y|X}$ is the conditional covariance matrix (Theorem 3.5.2). The partial correlation of Definition 3.6.1 is the correlation computed from $\Sigma_{Y|X}$. Chapter 6 shows that least squares estimates $\mathbf{B}_*$ in large samples, whether or not the regression function is linear. It also recovers the predictor as an orthogonal projection in the space of square-integrable random variables. For scalar Y , the coefficient vector used there is $\mathbf{B}_*^\top$.

Proposition 2.5.3 — Multiple correlation

Let Y be scalar with $\sigma_{YY} = \text{Var}(Y) > 0$, let Σ_{XX} be positive definite, and write $\sigma_{XY} = \text{Cov}(\mathbf{X}, Y)$. Over all $\mathbf{a} \neq \mathbf{0}$,

$$\max_{\mathbf{a}} \text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = \rho_{Y|X}^2 = \frac{\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}}{\sigma_{YY}},$$

attained at $\mathbf{a} = \Sigma_{XX}^{-1} \sigma_{XY}$, the coefficients of the best linear predictor (if $\sigma_{XY} = \mathbf{0}$, every $\mathbf{a}$ attains the maximum 0). Moreover $1 - \rho_{Y|X}^2 = \sigma_{Y|X} / \sigma_{YY}$.

Proof. $\text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = (\mathbf{a}^\top \sigma_{XY})^2 / (\sigma_{YY} \mathbf{a}^\top \Sigma_{XX} \mathbf{a})$. Write $\mathbf{a}^\top \sigma_{XY} = (\Sigma_{XX}^{1/2} \mathbf{a})^\top (\Sigma_{XX}^{-1/2} \sigma_{XY})$. The Cauchy–Schwarz inequality gives $(\mathbf{a}^\top \sigma_{XY})^2 \leq (\mathbf{a}^\top \Sigma_{XX} \mathbf{a})(\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY})$, with equality iff $\Sigma_{XX}^{1/2} \mathbf{a}$ is proportional to $\Sigma_{XX}^{-1/2} \sigma_{XY}$. Dividing gives the bound and the maximizer. The last identity is the definition of $\sigma_{Y|X} = \sigma_{YY} - \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}$, divided by σ_{YY} . ■

$\rho_{Y|X}^2$ is the population version of R^2 : the fraction of the variance of Y that the best linear predictor accounts for. Chapter 14 studies its sample version.

2.5.3 Conditioning: total expectation and total covariance

When the joint distribution is available, conditioning gives another decomposition of the covariance matrix. Conditional expectations and covariances are taken entry by entry, as in Definition 2.1.1, and $\text{Cov}(\mathbf{Y} | \mathbf{X}) = E[(\mathbf{Y} - E(\mathbf{Y} | \mathbf{X}))(\mathbf{Y} - E(\mathbf{Y} | \mathbf{X}))^\top | \mathbf{X}]$.

Proposition 2.5.4 — Laws of total expectation and covariance

If $\mathbf{Y}$ has finite second moments, then $E(\mathbf{Y}) = E[E(\mathbf{Y} | \mathbf{X})]$ and

$$\text{Cov}(\mathbf{Y}) = E[\text{Cov}(\mathbf{Y} | \mathbf{X})] + \text{Cov}(E(\mathbf{Y} | \mathbf{X})).$$

Proof. The first identity is the tower property of conditional expectation, applied to each component. For the second, let $\mathbf{m}(\mathbf{X}) = E(\mathbf{Y} | \mathbf{X})$ and write $\mathbf{Y} - \boldsymbol{\mu} = (\mathbf{Y} - \mathbf{m}(\mathbf{X})) +$

$(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})$. The outer product has two cross terms. The conditional expectation of $(\mathbf{Y} - \mathbf{m}(\mathbf{X}))(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top$ given $\mathbf{X}$ is $[\mathbf{E}(\mathbf{Y} | \mathbf{X}) - \mathbf{m}(\mathbf{X})](\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top = \mathbf{0}$, so its expectation is zero, and likewise for the other. The expectation of $(\mathbf{Y} - \mathbf{m})(\mathbf{Y} - \mathbf{m})^\top$ is $\mathbf{E}[\text{Cov}(\mathbf{Y} | \mathbf{X})]$ by the tower property, and that of $(\mathbf{m} - \boldsymbol{\mu})(\mathbf{m} - \boldsymbol{\mu})^\top$ is $\text{Cov}(\mathbf{m}(\mathbf{X}))$ because $\mathbf{E} \mathbf{m}(\mathbf{X}) = \boldsymbol{\mu}$. ■

Example. (A shared random effect)

Measurements $Y_1, \dots, Y_k$ on the same subject share a subject effect: $\mathbf{Y} = \alpha \mathbf{1} + \boldsymbol{\varepsilon}$, where α has variance τ^2 , $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_k$, and $\boldsymbol{\varepsilon}$ is independent of α with mean zero. Given α , the mean is $\alpha \mathbf{1}$ and the covariance is $\sigma^2 \mathbf{I}$. By [Proposition 2.5.4](#),

$$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_k + \tau^2 \mathbf{1} \mathbf{1}^\top,$$

so every pair of measurements has correlation $\tau^2/(\tau^2 + \sigma^2)$. Independent errors combined with a shared random component produce exactly this *equicorrelation* structure. It is the starting point of Chapter 32. Ordinary least squares ignores it, with consequences that Chapter 33 examines.

2.5.4 Exercises

2.5.4.1 A. Check your understanding

Exercise A1.

For scalar Y and X with $\text{Var}(X) > 0$, show that the best linear predictor has slope σ_{YX}/σ_{XX} , and that its mean squared error is $\sigma_{YY}(1 - \rho^2)$.

Exercise A2.

Let X be uniform on $(-1, 1)$ and $Y = X^2 + \varepsilon$, with ε independent of X , mean 0 and variance σ^2 . Find the best linear predictor of Y from X and its mean squared error, and compare with the mean squared error of $\mathbf{E}(Y | X)$.

2.5.4.2 B. Practice

Exercise B1.

Let (Y_1, Y_2, Y_3) have correlation matrix $\mathbf{R}$. Compute the partial covariance matrix of (Y_2, Y_3) given Y_1 from [Theorem 2.5.2](#), and show that the corresponding correlation is

$$\frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}}.$$

Exercise B2.

Let $\mathbf{X} = (X_1, X_1)^\top$ with $\text{Var}(X_1) = 1$, and let $\text{Cov}(Y, X_1) = c$. Find two different generalized inverses of Σ_{XX} that give different $\mathbf{B}_*$, and show that both give the same predictor $\mathbf{B}_*(\mathbf{X} - \mu_X)$.

Exercise B3.

A random label K takes values $1, \dots, g$ with probabilities π_k . Given $K = k$, $\mathbf{Y}$ has mean $\mathbf{m}_k$ and covariance Σ_k . Show that $\text{Cov}(\mathbf{Y}) = \sum_k \pi_k \Sigma_k + \sum_k \pi_k (\mathbf{m}_k - \bar{\mathbf{m}})(\mathbf{m}_k - \bar{\mathbf{m}})^\top$, where $\bar{\mathbf{m}} = \sum_k \pi_k \mathbf{m}_k$. Interpret the two terms as within-group and between-group covariance.

2.5.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{N}$ be the symmetric partitioned matrix

$$\mathbf{N} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

Show that $\mathbf{N}$ is nonnegative definite iff Σ_{XX} is nonnegative definite, $\mathcal{C}(\Sigma_{XY}) \subseteq \mathcal{C}(\Sigma_{XX})$, and $\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY}$ is nonnegative definite for some generalized inverse Σ_{XX}^- (equivalently, given the column-space condition, for every one). Show by a 2×2 example that the column-space condition cannot be dropped.

Solution. If the matrix is nonnegative definite, it is the covariance matrix of some $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$ (Theorem 2.2.3(b)). Then Σ_{XX} is a covariance matrix, the column condition is Theorem 2.5.1(c), and the Schur complement is $\text{Cov}(\mathbf{E})$ by Theorem 2.5.2(b). Conversely, the column condition gives $\Sigma_{YX} = \mathbf{K}\Sigma_{XX}$, and then $\Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY} = \mathbf{K}\Sigma_{XX}\mathbf{K}^\top$. With the nonsingular $\mathbf{T} = \begin{pmatrix} \mathbf{I} & -\mathbf{K} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$,

$$\mathbf{T} \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix} \mathbf{T}^\top = \begin{pmatrix} \Sigma_{YY} - \mathbf{K}\Sigma_{XX}\mathbf{K}^\top & \mathbf{0} \\ \mathbf{0} & \Sigma_{XX} \end{pmatrix},$$

which is nonnegative definite. So is the original matrix, since $\mathbf{a}^\top \mathbf{N} \mathbf{a} = \mathbf{b}^\top (\mathbf{T} \mathbf{N} \mathbf{T}^\top) \mathbf{b}$ with $\mathbf{b} = (\mathbf{T}^\top)^{-1} \mathbf{a}$. For the example, take $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Here $\Sigma_{XX} = 0$ is nonnegative definite, and with the generalized inverse 0 the Schur complement is $1 \geq 0$. But the determinant is -1 , so the matrix is not nonnegative definite. The column condition fails, because $\mathcal{C}(1) \not\subseteq \mathcal{C}(0)$. Without that condition the Schur complement also depends on the choice: every scalar t is a generalized inverse of 0, and the Schur complement is then $1 - 1 \cdot t \cdot 1 = 1 - t$, which is negative for $t > 1$. Under the column condition, $\Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY} = \mathbf{K}\Sigma_{XX}\mathbf{K}^\top$ for every choice, as shown above.